Kishor Joshi

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Professional Summary

Ph.D. student in Electrical and Computer Engineering with strong expertise in power electronics, motor drives, and control systems. Experienced in developing high-speed PMSM control algorithms, real-time embedded implementation, and simulation-based validation (HIL/SIL). Seeking internship opportunities in power electronics and advanced control systems.

Education

Ph.D. in Electrical and Computer Engineering | *Mississippi State University, Starkville, MS* **2023–Present**

- Current GPA: 4.0/**GTA2 Certified** Expected Graduation: July 2027
- Courses: Power Electronics, Power Electronic Applications, Feedback Control I and II, Nonlinear Control, Power Distribution Systems, Control of DER Systems, Optimal Control, Power System Optimization and Control, Power System Stability and Control, Computer Methods in Power Systems, Power Transmission Systems, High-Frequency Power Electronics, Alternative Energy Sources

Bachelor's in Electrical Engineering | *IOE Pulchowk Campus, Tribhuvan University, Lalitpur, Nepal* **2014–2018**

- Thesis: Implementation of PV Solar Inverter as STATCOM Percentage: 83%
- Courses: Electric Machine Design I & II, Electric Circuit I & II, Power Electronics, Electronic Circuits and Devices, Advanced Control Systems, Signal Processing, Microprocessors, Digital Logic Design, C Programming, and C++

Research Experience

Ph.D. Researcher 2023 – Present

Mississippi State University

- Developed advanced control strategies for ultra-high-speed PMSM drives operating up to 100,000 RPM
- Designed delay-aware MIMO control frameworks for stability enhancement in high-frequency systems, improving the speed margin by 188%
- Validated state-of-the-art and proposed methods with Hardware-in-the-Loop (HIL) using PLECS RT Box FPGA model in real time
- Implemented the control algorithms as embedded code on a Microprocessor and PMSM hardware, carrying designs from simulation onto physical hardware, achieving a maximum speed of 95,000 RPM

Professional Experience

Power Electronics Control Intern

Cummins Inc., Fridley, MN May 2025 – Aug 2025

- Designed and implemented ultra-high-speed PMSM drive control using FOC and DTC strategies for improved dynamic performance, considering ramp rate and hardware protection criteria to operate the compressor up to 60,000 RPM with full torque load
- Evaluated control robustness and system response through Hardware-in-the-Loop (HIL) and Software-in-the-Loop (SIL) validation
- Enabled real-time testing and rapid prototyping by integrating TI LaunchPad controller with FPGA-based HIL platform, using Speedgoat and Typhoon real-time simulators

Project Engineer

Prism International (DLRMP), Nepal 2019 – 2021

- Developed load flow and loss estimation tools using the Newton-Raphson method, enhancing the accuracy of system analysis
- Conducted system-level study of 20 substations to evaluate technical losses –distribution, transmission, and loading – to improve grid efficiency
- Prepared comprehensive technical reports and recommendations for Nepal Electricity Authority, supporting grid loss reduction initiatives in Nepal

Teaching Experience

Graduate Teaching Assistant

Mississippi State University 2023 – Present

- Delivered instructional support for undergraduate courses – Microprocessor, Foundation of Electrical and Computer Engineering – by preparing structured course materials and conducting guided problem-solving sessions
- Evaluated student performance through grading of labs, assignments, and examinations, ensuring timely and constructive feedback

- Mentored students to strengthen understanding of core engineering concepts, improving engagement and academic performance

Assistant Lecturer
Advanced College of Engineering and Management, Nepal 2022

- Delivered lectures in electrical circuits and control systems, enhancing student understanding through structured explanations and practical examples
- Supervised laboratory sessions in power electronics and electric machines, ensuring safe operation and effective hands-on learning
- Mentored students to develop strong analytical and problem-solving skills, improving their grasp of core engineering concepts

Selected Projects: Graduate

Ultra-High-Speed PMSM Drive Development

- Designed control strategies for an ultra-high-speed PMSM drive (up to 100k RPM) using FOC-based approaches on a TI C2000 (F28379D) LaunchPad
- Developed and analyzed MIMO control techniques to mitigate delay effects and improve high-speed stability and dynamic performance
- Built the complete drive system using a GaN-based inverter interfaced with a Maxon motor equipped with Hall sensor feedback
- Performed Hardware-in-the-Loop (HIL) testing using Plecs RT Box for real-time validation and rapid prototyping of control algorithms
- Utilized MATLAB, Plecs, and Code Composer Studio (CCS) for modeling, simulation, and embedded implementation

High-Speed Motor Control

- Developed and implemented a high-speed PMSM drive system using Direct Torque Control (DTC) with seamless open-loop to closed-loop transition strategy
- Designed and tuned PI controllers for closed-loop operation, achieving stable high-speed performance under varying load conditions
- Integrated a 200 kW XM3 inverter (450 V DC bus) with a fuel cell compressor PMSM, enabling efficient high-power operation
- Successfully validated the system in hardware, achieving motor speeds up to 60,000 RPM
- Engineered thermal management using active cooling to maintain system stability and prevent overheating during high-speed operation

Power Factor Correction (PFC)

- Developed and implemented a Power Factor Correction (PFC) control strategy in MATLAB/Simulink using enhanced Phase-Locked Loop (ePLL) for accurate grid synchronization
- Designed control architecture to regulate DC output voltage at 400 V while improving power factor and reducing input current harmonics
- Performed real-time Hardware-in-the-Loop (HIL) validation using Plecs RT Box, enabling rapid prototyping and verification of control performance
- Utilized MATLAB and Plecs for system modeling, simulation, and validation of control algorithms under dynamic operating conditions

Selected Projects: Under Graduate

PV-STATCOM

- Developed a dual-function inverter control enabling seamless operation as a grid-connected PV inverter under irradiance and as a STATCOM for reactive power compensation during low/no PV generation
- Designed switching and control logic to dynamically transition between active power injection and reactive power support modes
- Implemented and validated control algorithms in MATLAB/Simulink for varying irradiance and grid conditions

Wind-Solar Hybrid System

- Designed and modeled a hybrid renewable energy system integrating wind turbines, solar PV generation, and battery storage for reliable power supply
- Performed system-level analysis to optimize power sharing between wind, solar, and storage under varying load and environmental conditions
- Evaluated system performance and efficiency using MATLAB/Simulink, focusing on energy management and stability of the hybrid microgrid
- Proposed optimal configuration for rural electrification, improving system reliability and utilization of renewable resources

Controlled Rectifier

- Engineered and compared GTO- and thyristor-based controlled rectifiers for microgrid applications with isolated loads
- Investigated harmonic content, power factor variation, and switching characteristics under varying firing angles
- Analyzed performance trade-offs between converter topologies in terms of efficiency and output voltage quality

- Developed MATLAB/Simulink models to validate steady-state and transient response of rectifier systems

Transmission Line Cost Tool

- Developed an automated cost estimation tool using Excel/VBA for transmission line projects based on parameters such as voltage level, line length, and terrain conditions
- Designed calculation algorithms to evaluate material, installation, and infrastructure costs, improving estimation accuracy and efficiency
- Enabled rapid scenario analysis for different design configurations, supporting decision-making in planning and budgeting

Load Flow Analysis

- Developed and implemented Newton-Raphson load flow algorithms in Python and ETAP for power distribution system analysis
- Analyzed voltage profiles, real/reactive power flow, and system losses across multi-bus network configurations
- Evaluated system performance under varying load conditions, identifying voltage stability limits and loss distribution
- Validated simulation results across Python and ETAP platforms to ensure accuracy and reliability of power flow solutions

Technical Skills

Programming & Software: MATLAB/Simulink, PLECS, Python, C, C++, CCS, ETAP, Excel/VBA
Embedded Systems & Control: TI C2000, Real-Time Embedded Control, Hardware-in-the-Loop (HIL), Software-in-the-Loop (SIL)
Power Electronics: DC-DC Converters, Inverters, Power Factor Correction (PFC), PWM Techniques, GaN-based Converters
Motor Drives: PMSM Drives, Field-Oriented Control (FOC), Direct Torque Control (DTC), High-Speed Motor Drives, MIMO Control, DC Motor Control
Modeling & Simulation: MATLAB/Simulink, PLECS, Control System Design, Real-Time Simulation using RT Box
Power Systems: Load Flow Analysis, Distribution System Analysis, Renewable Energy Integration, Grid-Connected Systems
Hardware & Testing: Inverter Integration, Signal Measurement, Calibration, Debugging, Thermal Management, High-Power System Testing
Other Skills: Technical Documentation, Data Analysis, Problem Solving, Research and System-Level Design

Publications

[1] K. Joshi and M. Karimi-Ghartemani, "Systematic design and comparative analysis of two pi speed controllers for pmsm drives," in *2026 Southeast Symposium on Contemporary Engineering Topics (SSCET)*, 2026, pp. 1–6. DOI: 10.1109/SSCET69323.2026.11735668.

[2] K. Joshi, K. N. Tasnim, M. Moniruzzaman, M. Karimi-Ghartemani, and S. Choi, *A robust multivariable approach for current control of permanent magnet synchronous motor*, Under Review at IEEE Transactions on Industrial Electronics, 2026.

[3] M. Moniruzzaman, K. Joshi, M. R. Rahman, M. K. Islam, S. Choi, and M. K. Ghartemani, "Implementation and analysis of direct torque control on high-speed pmsms: A comparative study of commercial and laboratory-developed motors," in *2025 IEEE Applied Power Electronics Conference and Exposition (APEC)*, 2025, pp. 943–950. DOI: 10.1109/APEC48143.2025.10977544.

[4] I. Tamrakar, K. Joshi, P. Karki, S. Bimali, and T. Aryal, "Implementation of solar-pv inverter as statcom during nighttime in grid-connected system (pv-statcom)," *Journal of the Institute of Engineering*, vol. 15, no. 3, pp. 411–419, 2019. DOI: 10.3126/jie.v15i3.32762.

[5] K. B. Thapa, A. Maharjan, K. Kaphle, K. Joshi, and T. Aryal, "Paper modeling of wind-solar hybrid power system for off-grid in nepal and a case study," *Journal of the Institute of Engineering*, vol. 15, no. 3, pp. 360–367, 2019. DOI: 10.3126/jie.v15i3.32223.

Research Interests

High-Speed Electric Drives, Power Electronics, Advanced Motor Control, Delay-Aware Control Systems, Hardware-in-the-Loop Simulation, Renewable Energy Systems

Honors and Awards

- Bridge Funding – Bagley College of Engineering (2026)
- Full tuition waiver – Tribhuvan University (2014-2018)